

Hitender K. Oswal

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OBJECTIVE: to earn a Ph.D. in Computer Science for a career in machine learning in medical research.

EDUCATION

University of Washington, Allen School

Sept. 2023 – June 2027

B.S. in Computer Science (GPA 3.9)

- **Relevant Coursework:** Autonomous Robotics, Artificial Intelligence, Computational Biology, Neural Engineering, VR Systems, Advanced Machine Learning
- **Awards:** Dean's List (2023-Present), UW Edward and Theresa McMahon Scholar, Washington State Opportunity Baccalaureate Scholar, Undergraduate Research Conference Travel Award

DEEP LEARNING RESEARCH WORK AT KURLAB

KurtLab, University of Washington

May 2024 – Present

Researcher

Synthetic CT Generation for Stroke Lesion Segmentation

- In order to develop robust deep learning algorithms for automated stroke lesion segmentation in hemorrhagic and ischemic strokes using CT imaging, a sizable, diverse annotated dataset is required. A key aim of the project is to develop latent diffusion models which are able to generate accurate CT images from stroke lesion masks. These models will allow us to synthesize large-scale annotated CT datasets which can supplement existing datasets and improve performance of downstream models for diagnostic tasks by medical professionals.

Stroke Lesion Segmentation

ISLES'24 Challenge Winner

- Accurate identification of stroke lesion boundaries is critical for diagnosis and treatment. For the ISLES Challenge at MICCAI 2024, we applied clinically-informed preprocessing techniques to a stroke segmentation model, helping our team win first place. I have also explored how these preprocessing techniques can improve segmentation in low-resource settings and investigated explainable AI techniques from a Shapley perspective for multi-contrast medical image segmentation.

RESEARCH WORK AT AWS

Amazon Web Services

Summer 2026

Software Development Engineer Intern

- Designed an autonomous AI framework that extracts threats from cybersecurity intelligence, identifies security gaps, and generates validated system improvements.

Amazon Web Services

Summer 2025

Software Development Engineer Intern

- Designed a scalable microservice that actively monitors security threat intelligence and vends critical security data to customer accounts globally, requiring high throughput and reliability.

RESEARCH PUBLICATIONS

Medical Image Computing

- KurtLab. 2025. "Here Comes the Explanation: A Shapley Perspective on Multi-contrast Medical Image Segmentation." ***The 3rd World Conference on eXplainable AI (XAI 2025)***.
- KurtLab. 2025. "Clinically-Informed Preprocessing Improves Stroke Segmentation in Low-Resource Settings." ***MIRASOL Workshop, MICCAI 2025***.

- KurtLab. 2024. "An Ensemble Approach for Brain Tumor Segmentation and Synthesis." **MICCAI 2024**.
- KurtLab. 2024. (Challenge Winner) "How We Won the ISLES'24 Challenge by Preprocessing." **SWITCH Workshop, MICCAI 2024**.

Human-Computer Interaction

- Sushil K. Oswal and Hitender K. Oswal. 2026. "'It Was Frustrating to See These LLMs Glitch': Evaluating Their Workflows and Products for Accessible User Experiences by Novice BLV Users." **ICCHP 2026**.
- Sushil K. Oswal and Hitender K. Oswal. 2025. "Conversational AI for Accessible Website Design: Integrating LLM Assistants in Website Builders." **CWUAAT 2025, University of Cambridge/Springer**.
- Sushil K. Oswal and Hitender K. Oswal. 2024. "Examining the Accessibility of Generative AI Website Builder Tools for Blind and Low Vision Users: 21 Best Practices for Designers and Developers." **IEEE ProComm 2024**.
- Sushil K. Oswal and Hitender K. Oswal. 2023. "Study of a Smartphone App as a Bridge Assistive Technology for COVID-19 Home Test: 19 Essential Guidelines." **CWUAAT 2023, University of Cambridge/Springer**.
- Sushil K. Oswal and Hitender K. Oswal. 2023. "A Story of Four Documentation Designs for One Medical Device: User Experience Analysis and Recommendations." **ACM SIGDOC '23**.

Other Publications

- Hitender K. Oswal. 2024. "A Mentoring Model for Promoting Computational Thinking: Understanding Student Leadership and Learning Environment in an Extracurricular Robotics Team." **para.chi.dub 2024**.
- Hitender K. Oswal, et al. 2024. "Understanding the Leadership Structure and Mentoring Model of an Extracurricular Robotics Team: Key Findings from a Case Study." **ACM SIGCSE 2024**.

OTHER UW PROJECTS

UW Reality Lab, University of Washington

Nov. 2023 – Jun. 2024

Researcher

- Built an interactive Virtual Reality game that aids in the understanding of fundamental computer science concepts and data structures through visualization
- Awarded **Best Developer Tool Award** at DubHacks 2023

Extended Reality Association (XRA)

Sept. 2025 – Present

President

- Planned and hosted Hack the AM, the first Extended Reality Hackathon hosted at the University of Washington
- Orchestrated spatial computing initiatives and technical workshops for a community of 80 members, facilitating hands-on access to emerging hardware including Meta Quest 3 and Apple Vision Pro.

Composite Backdoors in Reasoning LLMs

April 2026 – June 2026

Project for Advanced Machine Learning

- Investigated literature related to backdoors in LLMs and designed a composite backdoor
- Tested the effectiveness of composite backdoors on modern reasoning LLMs by finetuning and evaluating 6 open-source LLMs

Adaptive Thermal Haptics for Virtual Reality

Jan. 2025 – May 2025

Built for VR Systems and VR Capstone

- Designed custom wireless, Peltier-driven wrist-mounted hardware to provide thermohaptic feedback
- Integrated real-time temperature feedback over Bluetooth in Unity for VR environments

SKILLS & MEMBERSHIPS

Skills: Python, Java, C#, PyTorch, TensorFlow, ROS, XR Development

Memberships: Association for Computing Machinery (ACM), IEEE, ACM Puget Sound SIGCHI Chapter